CHEMISTRY

1) Match the following Column-I with Column-II.

	Column-I		Column-II
(A)	Raoult's law	(P)	$\Delta T_f = K_f m$
(B)	Henry's law	(Q)	π = CST
(C)	Elevation of boiling point	(R)	$p = x_1 p_1^o + x_2 p_2^o$
(D)	Depression in freezing point	(S)	$\Delta T_b = K_b m$
(E)	Osmotic pressure	(T)	$p = K_H x$

(1) (A)–(S); (B)–(R); (C)–(Q); (D)–(P); (E)–(T)

(2) (A)–(R); (B)–(T); (C)–(S); (D)–(P); (E)–(Q)

(3) (A)–(T); (B)–(R); (C)–(Q); (D)–(P); (E)–(S)

(4) (A)–(S); (B)–(R); (C)–(T); (D)–(P); (E)–(Q)

2) ***Assertion :-*** Isotonic solutions don't show net phenomenon of osmosis.
Reason :- Isotonic solutions have equal osmotic pressure at constant temperature.

(1) Both assertion and reason are true and reason is the correct explanation of assertion.

(2) Both assertion and reason are true but reason is not the correct explanation of assertion.

(3) Assertion is true but reason is false.

(4) Both assertion and reason are false.

3) Given below are two statements : One is labelled as Assertion A and the other is labelled as Reason R :
Assertion : Raoult's Law is a special case of Henry's Law.
Reason: The partial vapour pressure of the species in solution is directly proportional to the mole fraction of the concerned species in solution.
In the light of the above statements, Choose the correct answer from the options given below :

(1) Both assertion and reason are false.

(2) Assertion is correct and reason is incorrect.

(3) Assertion is incorrect and reason is correct.

(4) Both assertion and reason are correct.

4) **Assertion :** Aquatic species are more comfortable in cold water in comparison to warm water.
Reason : At a given pressure, the solubility of oxygen gas in water increases with decrease in temperature

(1) Both assertion and reason are correct and reason is the correct explanation of assertion.

(2) Assertion is correct and reason is incorrect

(3) Assertion is incorrect and reason is correct

(4) Both assertion and reason are correct and reason is not the correct explanation of assertion

5) **Assertion :** People taking a lot of salty food experience the puffiness or swelling, called edema
Reason : There is water retention in tissue cells and intercellular spaces because of osmosis.

(1) Both assertion and reason are correct and reason is the correct explanation of assertion

(2) Both assertion and reason are correct and reason is not the correct explanation of assertion

(3) Assertion is correct and reason is incorrect

(4) Assertion is incorrect and reason is correct

6) Match the solute and solvent in column-I with examples in column-II and mark the appropriate choice :-

	Column-I		**Column-II**
(A)	Liquid in gas	(i)	Copper dissolved in gold
(B)	solid in solid	(ii)	Glucose dissolved in water
(C)	Liquid in solid	(iii)	Chloroform mixed with nitrogen gas
(D)	Solid in liquid	(iv)	Amalgam of mercury with sodium

(1) (A)→(i), (B)→(iii), (C)→(ii), (D)→(iv)

(2) (A)→(iii), (B)→(i), (C)→(iv), (D)→(ii)

(3) (A)→(ii), (B)→(iii), (C)→(iv), (D)→(i)

(4) (A)→(iii), (B)→(ii), (C)→(i), (D)→(iv)

7) Match list-I with list-II

	List-I		**List-II**
(A)	68% nitric acid and 32% water by mass	(I)	Minimum boiling azeotrope
(B)	Solution showing large positive deviation from Raoult's law may form	(II)	Ideal solution
(C)	n-hexane and n-heptane	(III)	Maximum boiling azeotrope
(D)	Two solutions having same osmotic pressure at a given temperature	(IV)	Isotonic solution

Choose the correct answer from the options given below :-

(1) (A)-I, (B)-III, (C)-II, (D)-IV

(2) (A)-III, (B)-IV, (C)-II, (D)-I

(3) (A)-III, (B)-I, (C)-IV, (D)-II

(4) (A)-III, (B)-I, (C)-II, (D)-IV

8)

1 g of a non-electrolyte solute dissolved in 50 g of benzene lowered the freezing point of benzene by 0.40 K. The freezing point depression constant of benzene is 5.12 K kg mol^{-1}. Find molar mass of the solute. (in g mol^{-1})

(1) 200

(2) 100

(3) 256

(4) 150

9) Normality and molarity of '15 V' H_2O_2 solution respectively is :

(1) 1.33 N, 2.66 M

(2) 1.33 M, 1.33 N

(3) 2.66 N, 1.33 M

(4) 3.99 N, 1.33 M

10) Two liquids A and B have P_A^o & P_B^o in the ratio of 1 : 3 and the ratio of number of moles of A & B in liquid phase are 1 : 3 then mole fraction of 'A' in vapour phase in equilibrium with the solution will be:

(1) 0.1

(2) 0.2

(3) 0.5

(4) 1.0

11) Molality of 10% (w/w) aqueous solution of glucose is :

(1) 0.31 m

(2) 0.62 m

(3) 0.55 m

(4) 1.00 m

12) K_H value for $O_2(g)$, $N_2(g)$, $H_2(g)$ and He(g) are 34.86, 76.48, 69.16 and 144.97 respectively. Arrange these gases in the order of their increasing solubility.

(1) $O_2 < H_2 < N_2 < He$

(2) $He < H_2 < N_2 < O_2$

(3) $H_2 < He < O_2 < N_2$

(4) $He < N_2 < H_2 < O_2$

13) The direction of osmosis can be reversed if a pressure _____ than the osmotic pressure is applied to the solution side having higher concentration.

(1) Smaller
(2) Equal
(3) Larger
(4) Larger or equal

14) Which one of the following is incorrect for ideal solution ?

(1) $\Delta P = (P_{observed} - P_{calculated\ by\ Raoult's\ law}) = 0$
(2) $\Delta H_{mix} = 0$
(3) $\Delta G_{mix} = 0$
(4) $\Delta S_{mix} > 0$

15) 1 mole of heptane (v. p = 92 mm of Hg) was mixed with 4 moles of octane (v. p = 31 mm of Hg). The vapour pressure of resulting ideal solution will be: (in mm of Hg)

(1) 93
(2) 43.2
(3) 30
(4) 28

16) Best solvent used to determine depression in freezing point is–

(1) Camphor
(2) Chloroform
(3) Benzene
(4) Water

17) If α is the degree of dissociation of Na_2SO_4. The Van't Hoff factor used for the calculation of molecular mass ?

(1) $i = 1 + 2\alpha$
(2) $i = 1 - 2\alpha$
(3) $i = 1 + \alpha$
(4) $i = 1 - \alpha$

18) The boiling point of C_6H_6, CH_3OH, $C_6H_5NH_2$ and $C_6H_5NO_2$ are 80°C, 65°C, 184°C and 212°C respectively. Which will show highest vapour pressure at room temperature:

(1) C_6H_6
(2) CH_3OH
(3) $C_6H_5NH_2$

(4) $C_6H_5NO_2$

19) A molal solution is one that contains one mole of a solute in :-

(1) 1000g of the solvent
(2) One litre of the solution
(3) One litre of the solvent
(4) 22.4 litres of the solution

20) What is the normality of 1M H_3PO_2 solution?

(1) 0.5N
(2) 1.0N
(3) 2.0N
(4) 3.0N

21) The Vant Hoff factor (i) for a dilute solution of $K_3[Fe(CN)_6]$ is :

(1) 10
(2) 4
(3) 5
(4) 0.25

22) Each pair forms ideal solution except ?

(1) C_2H_5Br and C_2H_5I
(2) C_6H_5Cl and C_6H_5Br
(3) C_6H_6 and $C_6H_5CH_3$
(4) C_2H_5I and C_2H_5OH

23) In osmosis phenomenon net flow of

(1) Solvent molecules occurs from higher concentration to lower concentration of solution.
(2) Solvent molecules occurs from lower concentration to higher concentration of solution.
(3) Solute molecules occurs from higher concentration to lower concentration of solution.
(4) Solute molecules occurs from lower concentration to higher concentration of solution.

24) Osmotic pressure of 0.585% w/v NaCl solution at 27°C is.

(1) 2.49 atm
(2) 5 atm
(3) 1.2 atm
(4) 3.8 atm

25) According to Henry's law the solubility of a gas in a liquid is expressed as $P = K_H X$.
where P = Partial pressure of gas

X = mole fraction of gas in the solution

K_H = Henry's law constant

the value of K_H for a particular gas will increased by

(1) Increasing the temperature

(2) Decreasing the temperature

(3) Increasing the pressure

(4) Decreasing the pressure

26) When 1.04g of $BaCl_2$ is present in 10^5g of solution, the concentration of solution is–

(1) 0.104 ppm

(2) 10.4 ppm

(3) 0.0104 ppm

(4) 104 ppm

27) The mixture that forms maximum boiling azeotropes:

(1) $C_2H_5OH + H_2O$

(2) $CH_3COCH_3 + CS_2$

(3) $C_7H_{16} + C_8H_{18}$

(4) $H_2O + HNO_3$

28) The vapour pressure of a given liquid will decrease if:

(1) Surface area of liquid is decreased

(2) The volume of liquid in the container is decreased

(3) The volume in the vapour phase is increased

(4) The temperature is decreased

29) An aqueous solution boils at 101°C. What is the freezing point of the same solution?
(Given : K_f = 1.86°C/m and K_b = 0.51°C/m)

(1) 3.647°C

(2) –3.647°C

(3) –0.364°C

(4) –3.647 K

30) The relationship between osmotic pressures (π_1, π_2 and π_3) at a definite temperature when 1g glucose, 1 g urea and 1 g sucrose are dissolved in 1 litre of water is (assume i = 1 for all) :-

(1) $\pi_1 > \pi_2 > \pi_3$

(2) $\pi_3 > \pi_1 > \pi_2$

(3) $\pi_2 > \pi_1 > \pi_3$

(4) $\pi_2 > \pi_3 > \pi_1$

31) A gaseous mixture contain 2 moles of $N_2(g)$ and 3 moles of $O_2(g)$ then mole fraction of $O_2(g)$ will be :-

(1) 0.5
(2) 0.2
(3) 0.4
(4) 0.6

32) During depression of freezing point in a solution the following are in equilibrium

(1) Liquid Solvent, Solid Solvent
(2) Liquid Solvent, Solid Solute
(3) Liquid Solute, Solid Solute
(4) Liquid Solute, Solid Solvent

33) The vapour pressure of pure liquid solvent is 0.50 atm. When a non-volatile solute is added to the solvent, its vapour pressure drops to 0.30 atm, thus mole fraction of the solute is

(1) 0.4
(2) 0.25
(3) 0.45
(4) 0.75

34) Assuming complete ionisation, the solution having maximum freezing point will be :-

(1) 1 M CaF_2
(2) 1.5 M $Al_2(SO_4)_3$
(3) 2 M NaCl
(4) 1 M $AgNO_3$

35) An azeotropic mixture of two liquids has a boiling point higher than boiling points of its components, then it :-

(1) Shows positive deviation from Raoult’s law
(2) Shows negative deviation from Raoult’s law
(3) Shows ideal behaviour
(4) is saturated

36) Which of the following concentration terms are temperature dependent :-

(1) M, m, N
(2) %(w/v), M,N
(3) F,m
(4) N, mole fraction

37) The molarity of 900 gm of water is :-

(1) 50M

(2) 55.5M

(3) 5M

(4) 5.55M

38) p_A and p_B are the vapour pressure of pure liquid components, A and B, respectively of an ideal binary solution. If x_A represents the mole fraction of component A, the total pressure of the solution will be.

(1) $p_B + x_A (p_B - p_A)$

(2) $p_B + x_A (p_A - p_B)$

(3) $p_A + x_A (p_B - p_A)$

(4) $p_A + x_A (p_A - p_B)$

39) Which statement about the composition of vapour over an ideal 1 : 1 molar mixture of benzene and toluene is correct ? Assume the temperature is constant at 25°C.
Vapour pressure data (25°C)

Benzene	75 mm Hg
Toluene	22 mm Hg

(1) The vapour will contain higher percentage of benzene

(2) The vapour will contain higher percentage of toluene

(3) The vapour will contain equal amount of benzene and toluene

(4) Not enough information is given to make a prediction

40) Raoult's law is obeyed by each constituent of a binary liquid solution when :-

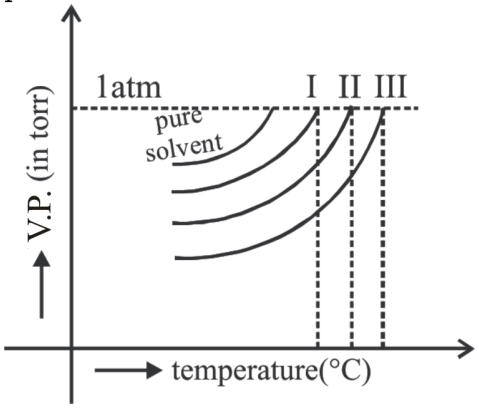
(1) the forces of attractions between like molecules are greater than those between unlike molecules

(2) the forces of attractions between like molecules are smaller than those between unlike molecules

(3) the forces of attractions between like molecules are identical with those between unlike molecules

(4) the volume occupied by unlike molecules are different.

41) The vapour pressure curves of the same solute in the same solvent are shown. The curves are parallel to each other and do not intersect. The concentrations of solutions are in order of :-



(1) I < II < III

(2) I = II = III

(3) I > II > III

(4) I > III > II

42) Which is not a colligative property ?

(1) Osmotic pressure

(2) Relative lowering in vapour pressure

(3) Depression in freezing point

(4) Refractive index

43) No. of moles of NaOH dissolved in 500 ml of 2M NaOH solution will be :-

(1) 1

(2) 2

(3) 3

(4) 4

44) 2500 ml aq. solution of urea contains 12×10^{24} molecules of urea. The molarity of the solution is :- (use $N_A = 6 \times 10^{23}$)

(1) 8M

(2) 4M

(3) 20M

(4) 12M

45) At a given temperature, total vapour pressure in Torr of a mixture of volatile components A and B is given by
$P = 120 - 75\ X_B$
hence, vapour pressure of pure A and B respectively (in Torr) are -

(1) 120, 75

(2) 120, 195

(3) 120, 45

(4) 75, 45

46) Which condition is true for both ideal and non ideal solution :-

(1) $\Delta H_{mixing} = 0$

(2) $\Delta S_{mixing} > 0$

(3) $\Delta G_{mixing} < 0$

(4) Both 2 & 3

47) On mixing 10 mL of toluene with 10 mL of benzene, the total volume of the solution is

(1) > 20 mL

(2) < 20 mL

(3) = 20 mL

(4) Cannot be predicted

48) Incorrect statement about azeotropes is:

(1) These are binary mixture having same composition in liquid and vapour phase

(2) They boil at a constant temperature

(3) The components of azeotropes can be separated by fractional distillation

(4) 68% nitric acid and 32% water by mass, form maximum boiling azeotrope

49) ___(A)____ injection are dissolved in water containing salts at particular ___(B)____ concentrations that matches ___(C)____ concentration. Here A, B and C refer to :

(1) Intravenous, ionic, blood plasma

(2) Intravenous, blood plasma, ionic

(3) Blood plasma, intravenous, ionic

(4) Blood plasma, ionic, intravenous

50) Compound A undergoes tetramerisation in the given solvent. The van't Hoff factor is

(1) 2.0

(2) 0.5

(3) 0.25

(4) 0.125

BIOLOGY

1)

Consider the following four statements (a-d) and select the option which includes all the correct ones :
(a) A multicarpellary apocarpous gynoecium is found in rose and lotus
(b) Pollen robbers consume pollen or nectar without bringing about pollination.
(c) The transformation of ovules into seed and ovary into fruit proceed simultaneously

(d) The pollen grain represents the male gametophyte.

(1) only a, b and c

(2) a, b, c and d

(3) only b, c and d

(4) only a, b and d

2)

Consider the following five statements (a-e) and select the option which includes all the correct ones only :-

(a) Vegetative cell is bigger, has abundant food reserve and a large irregularly shaped nucleus.
(b) Over 60 percent of angiosperms, pollen grains are shed at 3-celled stage.
(c) Pollen, grains can be stored for years in liquid nitrogen (–196°C).
(d) In members of Rosaceae, Leguminosae and Solanaceae the pollen grains are viable for months.

(e) Pollen grains are poor in nutrients.

(1) a, b, c, d

(2) a, c, d

(3) b, c, d

(4) a, c, d, e

3) Ploidy of MMC, nucellus, embryosac, megaspore and ovary wall are respectively :

(1) 2n, n, 7n, n, 2n

(2) 2n, 2n, 8n, n, n

(3) 2n, 2n, n, n, 2n

(4) n, 2n, n, 2n, n

4) What will be number of chromosomes in the endosperm of a seed formed by crossing of male plant (2N=40) and female plant (4N=28)

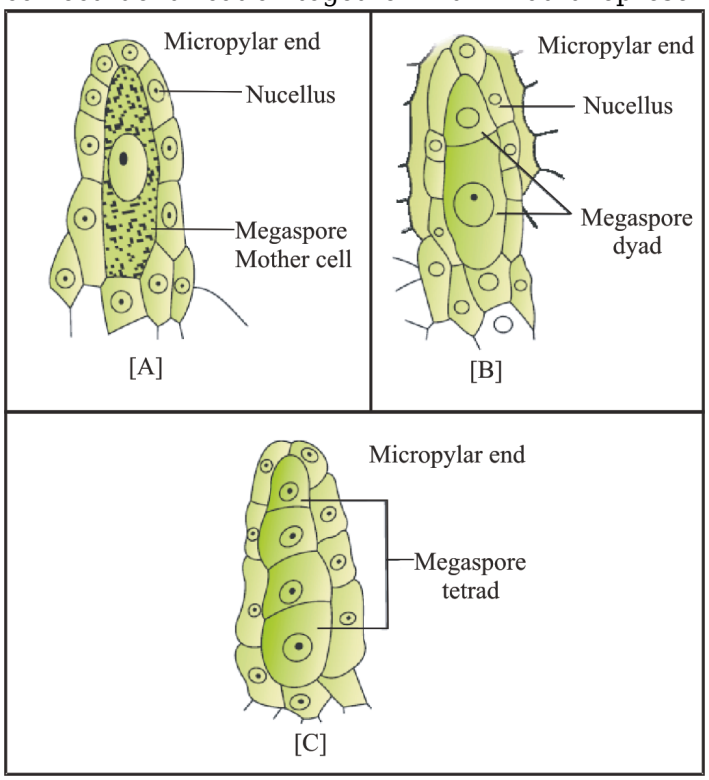
(1) 27

(2) 28

(3) 48

(4) 34

5) The figure below shows three steps [A] [B][C] of megasporogenesis. Select the option giving correct identification together with what it represents ?



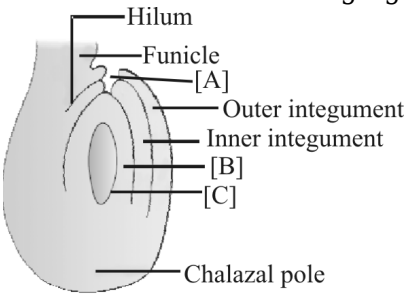
(1) B-Megaspore dyad is a resultant of meiosis-II

(2) C-Megaspore tetrad is a resultant of meiosis-I

(3) C-Megaspore tetrad is a resultant of meiosis-II

(4) A-Megaspore is a resultant of mitosis

6) A diagrammatic view of a typical anatropous ovule is shown in the following figure.Identify



labelled structures [A], [B] and [C] of given figure :-

(1) [A] Micropyle
[B] Filiform apparatus
[C] Nucellus

(2) [A] Micropyle
[B] Embryo
[C] Nucellus

(3) [A] Stigma
[B] Embryo
[C] Nucellus

(4) [A] Micropyle
[B] Nucellus
[C] Embryo-sac

7) Sequence of development during the formation of embryosac is –

(1) Archesporium → megaspore → megaspore mother cell

(2) Megaspore mother cell → archesporium → megaspore → embryosac

(3) Megaspore → megaspore mother cell → archesporium → embryosac

(4) Archesporium → megaspore mother cell → megaspore → embryosac

8)

Which group shows apocarpous gynoecium

(1) *Lotus*, Rose, *Michelia*

(2) Wheat, Chinarose, *Opium*

(3) *Hibiscus*, *Lotus*, *Wheat*

(4) *Papaver*, Lotus, *Hibiscus*

9) A. Two different plants are not necessarily required for sexual reproduction.
B. Chasmogamous flowers are invariably self pollinated.

(1) A & B both are false

(2) A is true, B is false

(3) A & B both are true

(4) A is false & B is true

10) Which one of the following options gives the correct categorisation of six plants according to the type of fruits (A, B, C) they have?

	A **FALSE** **FRUITS**	**B** **TRUE** **FRUITS**	**C** **PARTHENOCARPIC** **FRUITS**
(1)	Watermelon, Tomato	Apple, Strawberry	Banana, Grapes
(2)	Apple, Strawberry	Watermelon, Tomato	Banana, Grapes
(3)	Banana, Grapes	Watermelon, Tomato	Apple, Strawberry
(4)	Apple, Watermelon	Strawberry, Banana	Tomato, Grapes

(1) 1

(2) 2

(3) 3

(4) 4

11) Which one of following option gives the correct categorisation of six plants according to the type of pollination.

	(A) **Anemophily**	**(B)** **Hydrophily**	**(C)** **Entomophily**
(1)	Sea grass, maize	*Vallisneria, Pinus*	Cotton, *Eucalyptus*
(2)	Grass, maize	Sea grass *Vallisneria*	*Yucca*, cotton
(3)	*Eucalyptus*, maize	*Vallisneria pinus*	Sea grass, cotton
(4)	Sea grass, cotton	*Vallisneria, Eucalyptus*	*Pinus*, maize

(1) 1

(2) 2

(3) 3

(4) 4

12)

In Angiosperms, how many meiotic divisions are required to form 100 female gametophytes ?

(1) 400

(2) 300

(3) 200

(4) 100

13)

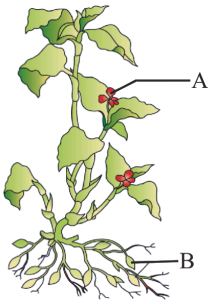
Monoecious plant prevents :-

(1) Autogamy

(2) Geitonogamy

(3) Both autogamy and geitonogamy

(4) Pollination is not possible

14) 

Select incorrect statement from the following for above diagram :-

(1) A – have exposed stigma and anther

(2) B – Always remain closed

(3) A – Invariably autogamous

(4) B – Found in *Viola*

15) How many meiotic divisions are required to produce 400 male gametes in angiosperms ?

(1) 10

(2) 30

(3) 50

(4) 100

16) If the number of chromosomes in male gametes of angiosperm is 24. The number of chromosomes in its endosperm, secondary nucleus and root would be respectively -

(1) 24, 48, 72

(2) 72, 48, 48

(3) 72, 24, 48

(4) 72, 24, 24

17) Which of the following statement is not correct?

(1) The generative cell divides meiotically to give rise two male gametes

(2) In over 60% angiosperms, pollination occurs in two celled stage

(3) Germ pore is the place where sporopollenin is absent

(4) Pollen grains are generally spherical measuring about 25-50 µm in diameter

18) Read the following statements carefully and identify the two correct statements ?
(A) The central cell after triple fusion becomes the primary endosperm cell and develops into the endosperm
(B) A dicot embryo consists of one embryonal axis and one cotyledon
(C) Radicle and root cap in monocot plant in embryonic stage enclosed in coleorrhiza

(D) The coconut water in tender coconut is cellular endosperm

(1) B and D
(2) A and C
(3) A and B
(4) A and D

19) Ploidy level of perisperm, synergids and middle layer is respectively:

(1) 3n, n, 2n
(2) 2n, n, 2n
(3) 2n, 2n, n
(4) n, 2n, n

20) Match the column-I with column-II :

	Column-I		**Column-II**
(A)	Funicle	I.	Mass of cells within ovule with more food.
(B)	Hilum	II.	Basal part of ovule
(C)	Integument	III.	One or two protective layers of ovule.
(D)	Chalaza	IV.	Region where body of ovule fuses with the funicle.
(E)	Nucellus	V.	Stalk of ovule.

(1) A-I; B-II; C-III; D-IV; E-V
(2) A-V; B-IV; C-III; D-II; E-I
(3) A-IV; B-II; C-I; D-III; E-V
(4) A-I; B-III; C-V; D-II; E-IV

21) Which of the following structure develops into female gametophyte ?

(1) Megaspore mother cell
(2) Functional microspore
(3) Functional megaspore
(4) Pollen mother cell

22) The nourishment of developing pollen grains is provided by :

(1) Epidermis
(2) Endothecium
(3) Middle layer
(4) Tapetum

23) In young anther a group of compactly arranged homogenous cells is called :

(1) Sporogenous tissue
(2) Microsporangium
(3) Pollen grains
(4) Tapetum

24) Many ovules are present in the ovary of :

(1) Wheat
(2) Rice (Paddy)
(3) Papaya
(4) Mango

25) Pollen pistil interaction involve all events from pollen deposition on the stigma to :

(1) Pollen tube formation
(2) Entry of pollen tube in to ovule
(3) Entry of pollen tube into embryosac
(4) Fertilization

26) Whorled phyllotaxy is present in:-

(1) *Alstonia*
(2) Guava
(3) Mustard
(4) *Calotropis*

27) Actinomorphic flower is not found in :-

(1) Mustard
(2) *Datura*
(3) Chilli
(4) Gulmohur

28) Which type of placentation is it called when the single ovule appears directly attached to base in a unilocular ovary?

(1) Superficial
(2) Free Central
(3) Basal
(4) Parietal

29) Which type of placentation is found in lemon ?

(1) Axile
(2) Parietal
(3) Free central

(4) Basal

30) Flowers are zygomorphic in :-

(1) Chilli
(2) *Cassia*
(3) *Dhatura*
(4) Canna

31) In flowers of pea there are five petals. In which two smallest anterior petals are called :-

(1) Standard
(2) Alae
(3) Keel
(4) Wing

32) Inferior ovary is present in :-

(1) Hypogynous flower
(2) Perigynous flowers
(3) Dichogamous flower
(4) Epigynous flower

33) In following, example of hypogynous flower is :-

(1) Mustard
(2) Chinarose
(3) Brinjal
(4) All of the above

34) Syngenesious condition is found in :-

(1) Chinarose
(2) Sunflower
(3) Tobacco
(4) Cucurbita

35) Presence of pappus is the characteristic of which fruit :-

(1) Caryopsis
(2) Cypsella
(3) Achene
(4) Samasa

36) In some fleshy fruits, endocarp is hard and stony due to the presence of stone cells. This type of fruits are found in which plant :-

(1) Mango and banana
(2) Mango and trapa
(3) Mango and coconut
(4) Coconut and tomato

37) Coconut fruit develops from :-

(1) Monocarpellary superior ovary
(2) Monocarpellary inferior ovary
(3) Multicarpellary superior ovary
(4) Multicarpellary inferior ovary

38) Endospermic seeds are found in :-

(1) Maize
(2) Pea
(3) Castor
(4) Both (1) and (3)

39) Medicinal plants like belladonna, *Ashwagandha* belongs to family :

(1) Solanaceae
(2) Liliaceae
(3) Cruciferae
(4) Fabaceae

40) Pinnately compound leaf is found in :-

(1) Peepal
(2) Neem
(3) Mango
(4) Silk cotton

41)

In which type of leaf, the leaflets are attached at a common point, i.e. at the tip of petiole :-

(1) Pinnately compound leaf
(2) Palmately compound leaf
(3) Both
(4) None

42) Epiphyllous condition is indicated by:

(1) $\overbrace{\text{C A}}$

(2) $\overbrace{\text{A}\ \ \text{G}}$

(3) $\overbrace{\text{K}\ \ \text{C}}$

(4) $\overbrace{\text{P}\ \ \text{A}}$

43) Which one of the following is not related to the floral formula of family brassicaceae :-

(1) $\underline{\text{G}}_{(2)}$

(2) $\text{A}_{(9)+1}$

(3) K_{2+2}

(4) C_{4}

44) Basal placentation is found in :-

(1) Compositae, poaceae

(2) Cruciferae, solanaceae

(3) Malvaceae, liliaceae

(4) Fabaceae, liliaceae

45)

Which of the following is a monocot family?

(1) Solanaceae

(2) Fabaceae

(3) Poaceae

(4) Cruciferae

46)  Above type of phyllotaxy can be seen in :-

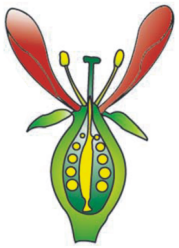
(1) Guava

(2) China Rose

(3) *Alstonia*

(4) Sunflower

47) Study the following diagram carefully and identify the position of ovary in this diagram :-



(1) Superior

(2) Inferior

(3) Half inferior

(4) Half superior

48) Identify the given diagram and select the correct option from the following table.

(a)  (b) 

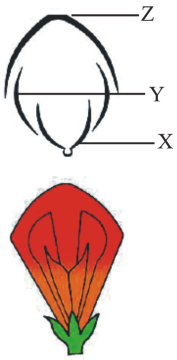
	(a)	(b)
(1)	China rose, lemon	Mustard, *Argemone*
(2)	*Dianthus, Primrose*	Chinarose, tomato
(3)	Mustard, *Argemone*	*Dianthus, Primrose*
(4)	Potato, lily	Grass, marigold

(1) 1

(2) 2

(3) 3

(4) 4



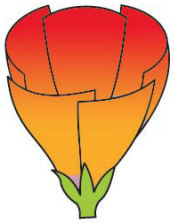
49) Identify "X" , "Y" and "Z" in given diagram ?

(1) X-Standard, Y-Wings, Z-Keel

(2) X-Wing, Y-Keel, Z-Standard

(3) X-Keel, Y-Wings, Z-Standard

(4) X-Keel, Y-Standard , Z-Wings



50) Following type of aestivation in petals can be seen in which plant :-

(1) *Calotropis*
(2) Cotton
(3) *Cassia*
(4) Mustard

ANSWER KEYS

CHEMISTRY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	1	4	1	1	2	4	3	3	1	2	4	3	3	2	1	1	2	1	2
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	2	4	2	2	1	2	4	4	2	3	4	1	1	4	2	2	2	2	1	3
Q.	41	42	43	44	45	46	47	48	49	50										
A.	1	4	1	1	3	4	3	3	1	3										

BIOLOGY

Q.	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70
A.	2	2	3	3	3	4	4	1	2	2	2	4	1	3	3	2	1	2	2	2
Q.	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	3	4	1	3	2	1	4	3	1	2	3	4	4	2	2	3	1	4	1	2
Q.	91	92	93	94	95	96	97	98	99	100										
A.	2	4	2	1	3	1	2	3	3	3										

SOLUTIONS

CHEMISTRY

1)

Raoult's law : $p = x_1 p_1^o + x_2 p_2^o$
Henry's law : $p = K_H x$
Elevation of boiling point: $\Delta T_b = K_b m$
Depression in freezing point: $\Delta T_f = K_f m$
Osmotic pressure: π = CST

2)

NCERT Pg. # 56

3) Ncert, Class12th , Part-1 , Article No: 1.4.2, Pg.No:12,Edition-2023-24.

4)

Ncert, Class 12th, part-1,
Article No: 1.3.2,
Pg No: 8, Edition - 2023-2024

5)

Ncert , Class12th , Part-1 , Article No: 1.6.4 , Pg.No:22,23, Edition-2023-24.

6)

Ncert , Class12th , Part-1 , Article No: 1.1 , Pg.No:2 , Edition-2023-24.

9) V = 11.2 × M
$M = \frac{V}{11.2} = \frac{15}{11.2} = 1.33$ M
N = M × v.f = 1.33 × 2
= 2.66 N

10) $\frac{Y_A}{Y_B} = \frac{P_A^o X_A}{P_B^o X_B} = \frac{1}{3} \times \frac{1}{3} = \frac{1}{9}$
$Y_A = 0.1$
$Y_B = 0.9$

11) 10% (w/w) means 10 gm glucose present in 100 gm solution.
Mass of solvent (water) = 100 – 10 = 90 gm
Moles of glucose = $\frac{10}{180}$ mole
Molality (m) = $\frac{10}{180} \times \frac{1000}{90}$
= 0.62 m

12)

NCERT XII Pg. # 40, Part - I
Higher the K_H value, lower is the solubility of the gas.

13)

NCERT-XII, Pg. # 57, Part-I

14) For solution formation; during mixing; $\Delta G_{mix} < 0$

15) V. P. of solution = $X_{heptane} \times 92 + X_{octane} \times 31$
= $\frac{1}{5} \times 92 + \frac{4}{5} \times 31 = 43.2$ mm of Hg

16) Kf of camphor is very high.

17) $i = 1 - \alpha + n\alpha$
$= 1 - \alpha + 3\alpha \Rightarrow 1 + 2\alpha$

18) Vapour pressure $\propto \frac{1}{\text{Boiling point}}$

19) $\text{Molality} = \frac{\text{Moles of solute}}{\text{Mass of solvent (in kg)}}$

20) N = M × n factor

21) $K_3[Fe(CN)_6] \rightarrow 3K^+ + [Fe(CN)_6]^{-3}$

22) C_2H_5OH shows hydrogen bonding as well as polarity both.

23)

Ncert, Class12th , Part-1 , Article No: 1.6.4 , Pg.No: 21 , Edition-2023-24.

24)

$\pi = iCRT$
(i = 2 for NaCl)
$\pi = \left(\frac{2 \times 0.585 \times 1000}{58.5 \times 100}\right) RT$ = 4.92 atm

25) NCERT-XII, Pg#39, para-5

26) $\frac{1.04}{10^5} \times 10^6$ = 10.4 ppm

27) $HNO_3 + H_2O$
will form negative deviation.

28)

$x_{(\square)} \underset{\text{Eq.}^m}{\rightleftharpoons} x_{(g)}$, K_p= Vp of liq, $\Delta_r H$ = +ve

T↓, Vp↓

29) $\frac{\Delta T_f}{1} = \frac{1.86}{.51}$
$\Delta T_f = 3.647$
Freezing point = –3.647°C

30) $\pi \propto$ molarity

31) $X_{O_2} = \frac{3\text{mol}}{2\text{mol} + 3\text{mol}} = \frac{3}{5} = 0.6$

32) During freezing of a solution only solvent freezes out and the equilibrium exists between solid and liquid forms of solvent

33) $P = X_{solvent} \times P°$
$X_{solvent} = \frac{0.3}{0.5} = 3/5$
$X_{solute} = 2/5$

34) Maximum freezing point for minimum ion.

35) Show negative deviation from Raoult's law.

36)

Concentration term including volume shows change on changing temperature.

37)

900 gm = 900 ml

Molarity = $\frac{900}{18 \times 0.9} \Rightarrow 55.5M$

38) $P_T = P_AX_A + P_BX_B$

$= P_AX_A + P_B(1 - X_A)$

$= P_AX_A + P_B - P_BX_A$

$P_T = P_B + X_A (P_A - P_B)$

39) $P_SY_A = P^0_A X_A \quad Y_A \alpha P^0_A$

Since benzene has higher vapour pressure thus it will have higher precentage of vapours.

40) Raoult's law is applicable for ideal solution.

41) Elevation in boiling point ∝ concentration of a solution. Thus, the order of concentration of solution is I < II < III.

42)

Refractive index is not a colligative property

43) No. of mole = $\frac{VM}{1000} = \frac{500 \times 2}{1000} = 1$

44) $\text{moles} = \frac{12 \times 10^{24}}{6 \times 10^{23}} = 20$

$v(\ell) = \frac{2500}{1000} = 2.5\ell$

$M = \frac{\text{moles}}{v(\ell)} = \frac{20}{2.5} = 8M$

45) $P = P^0_A x_A + P^0_B x_B$

$= P^0_A(1 - x_B) + P^0_B x_B$

$= P^0_A + (P^0_B - P^0_A)x_B$

$P = P^0_A - (P^0_A - P^0_B)x_B]$

Compare this equation with given equation in question.

46) For both ideal and non ideal solution, on mixing entropy increases (i.e. $\Delta S > 0$) and since it is a spontaneous process ($\Delta G < 0$)

47)

Because mixture of toluene and Benzene forms ideal solution.

48)

Azeotropes can't be seperated by fractional distillation.

49)

Intravenous injection are dissolved in water containing salts at particular ionic concentration that matches plasma concentration.

50) $\underset{1-\alpha}{4A} \rightleftharpoons \underset{\alpha/4}{A_4}$

$$i = \frac{1-\alpha+\frac{\alpha}{4}}{1} = 1-\frac{3}{4}\alpha$$

α = degree of association = 1 i.e. 100% association

$$i = 1-\frac{3}{4} = 0.25$$

BIOLOGY

51) NCERT (XII) Pg. # 6, 15, 20

52)

NCERT, XII Pg. # 7, 8

55)

NCERT-XII, Pg. # 10

56)

NCERT XII pg.# 9

57)

NCERT XII Pg # 9, 10

58)

NCERT XII Pg # 65

59)

NCERT XII Pg # 12

60) NCERT (XII) Pg # 20, 21

61)

NCERT XII Pg # 13, 14

63)

NCERT (XII) Pg. # 15

64)

NCERT–XII, Pg. # 12

67) NCERT (XII) Pg. # 7

68)

NCERT XII Pg. No. # 18, 19

70)

NCERT XII Pg # 9

71)

NCERT XII Pg # 9

72)

NCERT XII Pg # 5

73)

NCERT XII Pg # 5

74)

NCERT-XII, Pg # 9

75)

NCERT XII Pg # 17

76) NCERT XI Pg. # 61

77) NCERT XI Pg. # 62

78) NCERT XI Pg.# 65

79) NCERT (XI) Pg. # 65

80) NCERT XI Pg.# 62

81) NCERT (XI) Pg. # 64

82)

NCERT XI Pg # 62, 63

83) NCERT XI Pg. # 63

86) NCERT-XI Pg.# 65, 66

87) NCERT XI Pg # 65

88) NCERT (XI) Pg. # 66, 67

89) NCERT (XI) Pg. # 69

90)

NCERT XI Pg # 61

91)

NCERT XI Page No. 60, 61

93)

NCERT XI Pg # 67

96) NCERT-XI Page No. 61

97) NCERT (XIth) Pg. # 62

98) NCERT XI Pg.# 65

99) NCERT XI Pg. # 64

100) NCERT (XI) Pg. # 64